

Timeline Check In

Macey Hartmann and Erick Morales Oyola

2026-05-22

Setup

Packages

We loaded packages for data wrangling, file organization, missing-data exploration, and visualization. The main workflow uses `tidyverse` for cleaning and summarizing vegetation data, `here` for file paths, and plotting packages for exploratory figures.

Data

We used two vegetation datasets. The main dataset, `veg.csv`, contains vegetation survey observations, including species identity, percent cover, cover category, habitat type, transect, site, and year. The metadata file, `vp_veg_metadata.csv`, provides supporting information about vegetation transects, including the number of quadrats sampled within each transect-year.

Ideas for analysis and background

Our project focuses on salt marsh vegetation at North Campus Open Space (NCOS), with the goal of understanding which plant species characterize this restored wetland habitat and how those patterns have changed across survey years. Rather than treating vegetation cover as only a set of numbers, we are using percent cover to describe how the salt marsh is developing as a plant community over time. In this sense, our project asks how restoration becomes visible through the species that take root, persist, expand, or decline across the marsh.

Salt marsh vegetation is especially useful for understanding restoration trajectory because these plant communities are shaped by strong environmental filters, including salinity, tidal flooding, saturated soils, and seasonal changes in water availability. In southern California salt

marshes, native halophytes such as pickleweed (*Salicornia pacifica*) and saltgrass (*Distichlis spicata*) often help define the structure and identity of the habitat (Ferren 1985; North Campus Open Space 2025). Because these species are adapted to stressful coastal wetland conditions, changes in their mean percent cover can help indicate whether the restored salt marsh is developing toward a recognizable native marsh community.

This question is particularly relevant at NCOS because the site was restored from the former Ocean Meadows Golf Course into a mosaic of wetland and upland habitats, including salt marsh, brackish marsh, seasonal wetlands, and transitional habitats (Cheadle Center for Biodiversity and Ecological Restoration and Environmental Science Associates 2016; Rickard 2024). Broader NCOS monitoring reports evaluate restoration progress across habitat types, while our project narrows in on salt marsh vegetation specifically. By focusing on dominant species and native/non-native cover through time, we can connect the survey data to larger questions about restoration progress, habitat identity, and community development.

Previous work on California wetland restoration has shown that vegetation establishment can be slow, variable, and strongly shaped by local site conditions such as elevation, soil salinity, soil moisture, planting, and adaptive management (Beheshti et al. 2023). This matters for our analysis because shifts in dominant plant cover are not just visual patterns in a graph. They may reflect ecological processes such as native plant establishment, non-native plant expansion, hydrologic change, disturbance, or interannual variability in rainfall and growing conditions.

Ecological background

Salt marshes are dynamic coastal wetlands where plant communities exist at the edge of land and water. Their vegetation patterns are shaped by both physical stress and biological interactions, making them useful indicators of environmental conditions and restoration progress. In southern California, seasonal rainfall and dry summers can influence freshwater input, soil moisture, salinity, and plant growth, which may affect how vegetation cover changes from year to year (Ferren 1985).

At NCOS, we expect dominant salt marsh species to include native salt-tolerant plants such as *Salicornia pacifica* and *Distichlis spicata*. The NCOS coastal salt marsh habitat page describes pickleweed as usually the dominant species in coastal salt marshes and also identifies saltgrass, alkali heath, fleshy jaumea, and California saltbush as characteristic salt marsh plants (North Campus Open Space 2025). These expectations give us a useful ecological reference point for interpreting our data: if the restored salt marsh is developing toward a recognizable coastal salt marsh community, we would expect these native halophytes to contribute strongly to vegetation cover.

Native and non-native cover are also important to examine because restoration success depends not only on whether plants are present, but on which plants are shaping the habitat. A marsh with high total vegetation cover may still be ecologically different depending on whether that

cover is dominated by native salt marsh species or by non-native plants. NCOS monitoring reports describe vegetation cover, native biodiversity, and relative native cover as important indicators of restoration progress (Rickard 2024). Our project builds from that monitoring context by focusing more closely on the species-level patterns that contribute to salt marsh identity through time.

Planned analyses

Our first visualization will summarize which plant species most strongly characterize salt marsh and closely related brackish marsh vegetation at NCOS. To do this, we will summarize mean percent cover for each plant species across survey years and transects, then rank species from highest to lowest mean cover. We will use the 90th percentile of mean percent cover as a cutoff for identifying dominant species, which gives us a clearer mathematical basis than selecting species only by visual inspection.

Our second analysis will examine how dominant species cover changes across survey years. After identifying species at or above the 90th percentile of mean percent cover across the full filtered dataset, we will summarize their mean percent cover by year and visualize them with a stacked bar plot. In this figure, the y-axis represents the summed mean percent cover of the dominant species in each survey year, so bar height shows how much cover those high-contributing species represent through time.

Our third analysis will focus specifically on non-native species in the filtered salt marsh/brackish marsh dataset. Rather than only comparing total native and non-native cover, we will identify the five non-native species with the highest mean percent cover across all survey years, then summarize their mean percent cover by year. We will visualize these species with a stacked bar plot so we can see whether non-native cover is being driven by the same species through time or whether different non-native plants become more important in different years. This analysis is useful because a simple native/non-native proportion plot can show the relative amount of non-native cover, but it does not show which species are responsible for that pattern. By focusing on the top non-native species, we can connect non-native cover to actual plant identities and better interpret whether non-native vegetation reflects a consistent community pattern, a temporary increase in a few species, or a shift in the plants contributing to marsh composition.

Our fourth analysis will provide broader context by examining total vegetation cover across survey years. Unlike the dominant-species figure, this analysis includes all native and non-native plant species in the filtered salt marsh/brackish marsh dataset. We will summarize total mean vegetation cover by year and visualize it as a line plot with standard error, which will help us interpret whether changes in species composition are happening alongside broader increases or decreases in overall vegetation cover.

Together, these visualizations connect our figures to the main project questions. The species-rank plot identifies the plants that most strongly characterize the habitat, the dominant-species

plot shows how the top five species change through time, the proportional native/non-native plot shows how plant origin contributes to habitat identity, and the dominant non-native species plot shows which non-native plants are driving non-native cover across survey years. The total-cover plot then provides broader context for overall vegetation development across the monitoring period.

Data cleaning and wrangling

We filtered the vegetation dataset to focus on salt marsh and closely related brackish marsh vegetation, then limited the surveys to August and September because late summer and early fall are when vegetation cover is expected to be highest and most comparable across years. We used `complete()` to make implicit absences explicit as zero percent cover so that species not recorded in a given transect-year are treated as absent rather than missing (this matters for calculating accurate transect-level means). We then joined the vegetation data with the meta-data file so we could account for the number of quadrats sampled in each transect-year when computing mean cover. Finally, we filtered to native and non-native plant cover categories so our analyses focused on vegetation rather than bare ground, thatch, or other non-plant cover types.

```
saltmarsh <- veg |>
  # Filtering to salt marsh and brackish marsh because these habitat types
  ↪ share
  # similar plant communities and together represent the core restored
  ↪ wetland area at NCOS
  filter(str_detect(habitat, "Salt Marsh") | str_detect(habitat, "Brackish
  ↪ Marsh")) |>
  # Restricting to August and September; late summer is when cover peaks and
  ↪ is most stable across years, reducing seasonal noise in comparisons
  mutate(date = as.Date(date),
         month = month(date)) |>
  filter(month %in% c(8, 9))
```

```
saltmarsh_complete <- saltmarsh |>
  # Using complete() to make implicit absences explicit: if a species was not
  # recorded in a transect-year, it is assigned zero percent cover rather
  ↪ than
  # being dropped from the dataset, which would inflate mean cover estimates
  complete(year, transect_name, psoc, transect_distance,
         fill = list(percent_cover = 0)) |>
  group_by(year, transect_name, psoc, cover_category) |>
  summarize(sum_pc = sum(percent_cover, na.rm = TRUE)) |>
```

```

ungroup() |>
# Creating a composite key to join vegetation data with quadrat metadata
unite("year_pool", year, transect_name, remove = FALSE) |>
left_join(metadata, by = "year_pool") |>
select(-year.y, -transect_name.y) |>
rename(year = year.x,
       transect_name = transect_name.x) |>
# Dividing summed cover by the number of quadrats to get mean percent cover
# per quadrat within each transect-year, which accounts for variation in
# sampling effort across transects and years
mutate(mean_pc = sum_pc / num_quad)

```

```

saltmarsh_plants <- saltmarsh_complete |>
# Keeping only native and non-native plant cover; removing bare ground,
↪ thatch,
# and other non-plant categories so all analyses focus on vegetation
filter(cover_category %in% c("NATIVE COVER", "NON-NATIVE COVER")) |>
# Replacing underscores with spaces in species codes so names render
↪ cleanly
# in figure axes and legends without extra formatting steps
mutate(species_label = str_replace_all(psoc, "_", " "))

```

Planned visualizations and descriptive analyses

For this stage of the project, our work focuses mainly on descriptive analyses and visualizations that help us understand salt marsh/brackish marsh vegetation patterns at NCOS. These summaries allow us to identify dominant species, compare native and non-native cover, and explore how vegetation cover changes across survey years. For the final project, we plan to add a simple statistical analysis using linear models to test whether vegetation cover changes through time.

Our planned statistical analysis will use survey year as the explanatory variable and mean percent cover as the response variable. We will first test whether total mean vegetation cover changes across years, then consider separate statistical tests for native cover, non-native cover, or individual dominant species if the data structure supports those comparisons. These tests will help us move beyond visual interpretation by asking whether the patterns we see in the figures suggest directional changes in vegetation cover over the monitoring period.

```

# Creating a species-level summary table for the rank plot
species_rank <- saltmarsh_plants |>
# Grouping by species and cover category so each species keeps its
↪ native/non-native status

```

```

group_by(psoc, species_label, cover_category) |>
# Calculating mean percent cover, standard error, and sample size for each
  ↳ species
summarize(mean_pc = mean(mean_pc, na.rm = TRUE),
          se_pc = sd(mean_pc, na.rm = TRUE) / sqrt(n()),
          n = n()) |>
# Removing grouping so later steps do not accidentally stay grouped
ungroup() |>
# Ordering species from highest to lowest mean percent cover
arrange(desc(mean_pc))

# Calculating the 90th percentile of species mean percent cover as the
  ↳ dominance threshold.
# We chose the 90th percentile rather than a visual cutoff because it
  ↳ provides a
# reproducible, data-driven criterion: only species whose average cover
  ↳ exceeds
# 90% of all recorded species are classified as dominant. This threshold
  ↳ identifies
# the highest-contributing species without arbitrarily inflating or deflating
  ↳ the set.
dominance_cutoff <- quantile(species_rank$mean_pc,
                             probs = 0.90,
                             na.rm = TRUE)

# Identifying dominant species as those at or above the 90th percentile
dominant_species <- species_rank |>
  filter(mean_pc >= dominance_cutoff) |>
  pull(psoc)

# Creating a table that ranks dominant species from highest to lowest mean
  ↳ cover
# for use in ordering the stacked bar plot legend and stack order
dominant_species_order <- species_rank |>
  filter(psoc %in% dominant_species) |>
  arrange(desc(mean_pc)) |>
  select(psoc, species_label, overall_mean_pc = mean_pc)

# Summarizing dominant species cover by year for Figure 1
dominant_by_year <- saltmarsh_plants |>
  # Keeping only species at or above the 90th percentile cutoff
  filter(psoc %in% dominant_species) |>

```

```

# Grouping by year, transect, species, and cover category
group_by(year, transect_name, psoc, species_label, cover_category) |>
# Calculating species cover within each transect-year
summarize(sum_pc = sum(mean_pc, na.rm = TRUE)) |>
ungroup() |>
# Averaging across transects within each year and species so that
# bar height reflects mean cover rather than summed cover across transects
group_by(year, psoc, species_label, cover_category) |>
summarize(mean_pc = mean(sum_pc, na.rm = TRUE)) |>
ungroup() |>
# Joining the overall species rank back into the plotting dataset
left_join(dominant_species_order, by = c("psoc", "species_label")) |>
# Ordering species so the highest-cover species appears at the bottom of
  ↳ the stacked bar (most visually stable position)
mutate(species_label = factor(species_label,
                             levels =
                               ↳ rev(dominant_species_order$species_label)))

# Keeping the top 20 species for the rank plot, excluding singletons (n > 1)
# to avoid species with only one observation inflating apparent cover
species_rank_top20 <- species_rank |>
  filter(n > 1) |>
  slice_max(mean_pc, n = 20)

# Creating a color palette for dominant species; green shades for natives,
# coral/red shades for non-natives, to make origin immediately readable
species_rank_palette <- c(
  # Native species
  "Salicornia pacifica"      = "#00441B",
  "Distichlis spicata"      = "#238B45",
  "Atriplex lentiformis"    = "#74C476",
  "Distichlis littoralis"   = "#006D2C",
  "Suaeda taxifolia"        = "#41AB5D",
  "Eleocharis macrostachya" = "#C4E8D9",
  "Elymus triticoides"      = "#A1D99B",
  "Isocoma menziesii"       = "#2CA25F",
  "Frankenia salina"        = "#99D8C9",

  # Non-native species
  "Raphanus sativus"        = "#D85A30",
  "Festuca perennis"        = "#B8322A"
)

```

```

# Creating transect-level native and non-native cover totals for underlying
  ↪ points in Figure 2
# Each row represents the total native or non-native cover within one
  ↪ transect-year,
# which is what we plot as raw observations behind the yearly means
native_nonnative_points <- saltmarsh_plants |>
  # Grouping by year, transect, and cover category to keep transect-level
  ↪ observations
  group_by(year, transect_name, cover_category) |>
  # Summing native or non-native cover within each transect-year
  summarize(total_pc = sum(mean_pc, na.rm = TRUE)) |>
  ungroup()

# Summarizing to yearly means and standard errors for the line and ribbon in
  ↪ Figure 2
native_nonnative_raw <- saltmarsh_plants |>
  # First collapsing to transect-level totals so each transect-year
  # contributes one observation per cover category to the yearly mean
  group_by(year, transect_name, cover_category) |>
  summarize(total_pc = sum(mean_pc, na.rm = TRUE)) |>
  ungroup() |>
  # Then averaging across transects within each year and cover category
  group_by(year, cover_category) |>
  # Calculating mean cover and standard error for each cover category in each
  ↪ year
  summarize(mean_cover = mean(total_pc, na.rm = TRUE),
            se_cover = sd(total_pc, na.rm = TRUE) / sqrt(n())) |>
  ungroup()

# Identifying the five non-native species with the highest average cover
  ↪ across all years;
# focusing on the top five keeps the stacked bar readable while capturing the
  ↪ species
# most responsible for non-native cover patterns in the marsh
top_nonnative_species <- saltmarsh_plants |>
  filter(cover_category == "NON-NATIVE COVER") |>
  group_by(psoc) |>
  summarize(mean_pc = mean(mean_pc, na.rm = TRUE)) |>
  ungroup() |>
  slice_max(mean_pc, n = 5) |>
  pull(psoc)

```



```

# Summarizing top non-native species cover by year for Figure 3
top_nonnative_by_year <- saltmarsh_plants |>
  filter(cover_category == "NON-NATIVE COVER",
         psoc %in% top_nonnative_species) |>
  mutate(species_label = str_replace_all(psoc, "_", " ")) |>
  # First summarizing within transect-years
  group_by(year, transect_name, psoc, species_label) |>
  summarize(sum_pc = sum(mean_pc, na.rm = TRUE)) |>
  ungroup() |>
  # Then averaging across transects within each year
  group_by(year, psoc, species_label) |>
  summarize(mean_pc = mean(sum_pc, na.rm = TRUE)) |>
  ungroup()

# Creating transect-level total vegetation cover values for the underlying
# raw data points in Figure 5; each row is one transect-year observation
total_cover_points <- saltmarsh_plants |>
  group_by(year, transect_name) |>
  summarize(total_pc = sum(mean_pc, na.rm = TRUE)) |>
  ungroup()

# Summarizing total vegetation cover by year for the line and ribbon in
# Figure 5
total_cover <- total_cover_points |>
  group_by(year) |>
  summarize(mean_total = mean(total_pc, na.rm = TRUE),
            se_total = sd(total_pc, na.rm = TRUE) / sqrt(n())) |>
  ungroup()

```

Visualizations directly relevant to answering questions

The following four figures directly address our project questions by connecting plant identity, cover, and origin through time. Figure 1 focuses on the dominant plant species overall, Figure 2 compares the contribution of native and non-native cover across years, Figure 3 examines which non-native species contribute most to non-native cover, and Figure 4 ranks the top species by mean cover to justify our dominant-species cutoff.

Figure 1: Dominant species percent cover across survey years

```

# Creating Figure 1: dominant species cover across survey years as a stacked
  ↪ bar plot.
# Stacking allows the total dominant-species cover and individual species
  ↪ contributions
# to be read from the same figure.
fig1 <- ggplot(data = dominant_by_year,
               aes(x = year,
                   y = mean_pc,
                   fill = species_label,
                   group = species_label)) +
  # Stacking bars so each color segment shows one dominant species'
  ↪ contribution
  geom_col(position = "stack") +
  # Applying custom species colors; breaks argument controls legend order
  # so it matches the stacking order (highest cover at bottom)
  scale_fill_manual(values = species_rank_palette,
                    breaks = dominant_species_order$species_label) +
  # Showing each survey year as its own x-axis label
  scale_x_continuous(breaks = seq(2018, 2025, 1)) +
  # Adding axis labels and legend title
  labs(x = "Survey year",
       y = "Mean percent cover (%)",
       fill = "Species") +
  # Using a clean theme for readability
  theme_minimal() +
  # Formatting the legend and italicizing species names
  theme(legend.position = "bottom",
        legend.text = element_text(face = "italic"),
        legend.title = element_text(face = "bold"))

# Displaying Figure 1
fig1

```

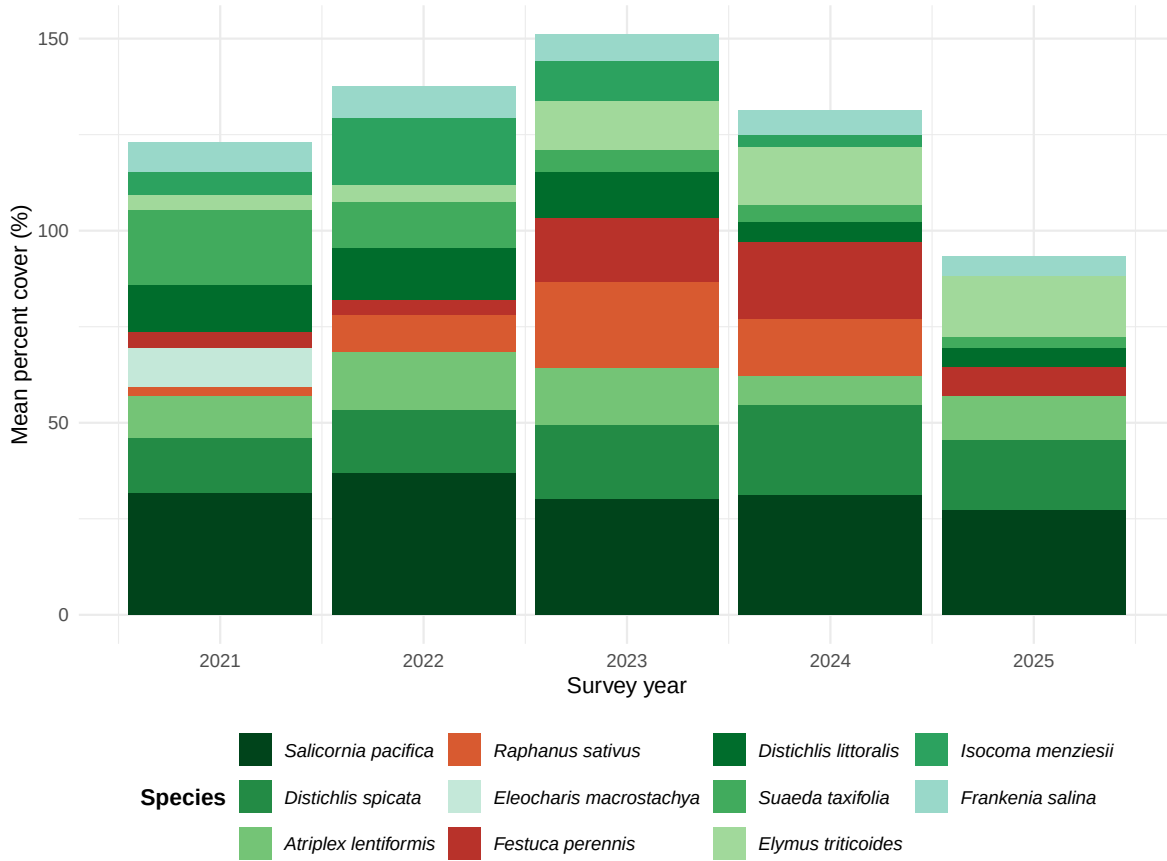


Figure 1: Mean percent cover of dominant salt marsh/brackish marsh plant species at NCOS across survey years, averaged across transects within each year. Dominant species were identified as species at or above the 90th percentile of mean percent cover across all survey years. Bar height reflects total mean cover of the dominant species and varies across years. Green shades indicate native species; coral indicates non-native species.

The original figure was unclear about what the y-axis represented, so we revised the wrangling approach to make the summary more interpretable. Rather than averaging directly across all observations, we first calculated transect-level totals for each dominant species within each year, then averaged those values across transects. This means the y-axis represents mean percent cover averaged across transects within each survey year, and bar height varies across years based on the summed cover of the dominant species. Dominant species were identified using the 90th percentile of mean percent cover across the full filtered dataset, rather than by visual inspection. The “Other” category was removed so the figure focuses only on species that met the dominant-species cutoff, and species are stacked within each year from highest to lowest mean cover.

Figure 2: Native and non-native cover across survey years

```
# Creating Figure 2: native and non-native vegetation cover over time.
# Using a line plot rather than stacked bars to emphasize temporal trend and
# allow both categories to share the same axis for direct comparison.

# First, creating a label dataset with the last year's values for direct
  ↪ labeling
fig2_labels <- native_nonnative_raw |>
  group_by(cover_category) |>
  filter(year == max(year)) |>
  ungroup() |>
  # Cleaning up the cover category text for display
  mutate(label = str_to_title(str_replace(cover_category, " COVER", "")))

fig2 <- ggplot(data = native_nonnative_raw,
               aes(x = year,
                   y = mean_cover,
                   color = cover_category,
                   fill = cover_category)) +
  # Faint jittered points show the underlying transect-level observations;
  # each point is one transect-year total for that cover category
  geom_point(data = native_nonnative_points,
             aes(x = year,
                 y = total_pc,
                 color = cover_category),
             inherit.aes = FALSE,
             alpha = 0.35,
             size = 1.8,
             position = position_jitter(width = 0.08, height = 0)) +
  # Shaded ribbons show ± 1 SE around each yearly mean
  geom_ribbon(aes(ymin = mean_cover - se_cover,
                 ymax = mean_cover + se_cover),
             alpha = 0.2,
             color = NA) +
  # Lines connect yearly means to show temporal trend
  geom_line(linewidth = 1) +
  # Larger points mark the yearly mean values
  geom_point(size = 3) +
  # Direct labels at the end of each line using ggrepel to avoid overlap
  geom_label_repel(data = fig2_labels,
                  aes(x = year,
                      y = mean_cover,
```

```

        label = label,
        color = cover_category),
    inherit.aes = FALSE,
    nudge_x = 0.3,
    hjust = 0,
    size = 3.5,
    fontface = "bold",
    show.legend = FALSE) +
scale_color_manual(values = c("NATIVE COVER" = "#1D9E75",
                              "NON-NATIVE COVER" = "#D85A30")) +
scale_fill_manual(values = c("NATIVE COVER" = "#1D9E75",
                             "NON-NATIVE COVER" = "#D85A30")) +
scale_x_continuous(breaks = seq(2018, 2025, 1),
                   # Expanding right side slightly to make room for direct
                   ↪ labels
                   expand = expansion(mult = c(0.02, 0.15))) +
labs(x = "Survey year",
     y = "Mean percent cover ( $\pm$  1 SE)") +
theme_minimal() +
# Removing legend since cover categories are labeled directly on the figure
theme(legend.position = "none")

# Displaying figure 2
fig2

```



Figure 2: Mean percent cover (± 1 SE) of native and non-native vegetation in salt marsh habitat at NCOS across survey years. Faint points show transect-level observations (one point per transect-year per cover category); lines connect yearly means; shaded bands represent ± 1 standard error. Cover categories are labeled directly on the figure. Native cover is shown in green; non-native cover in coral.

The original proportional stacked bar was nearly identical to panel (b) of figure 9 in the NCOS monitoring report, making it redundant. We replaced it with a line plot showing raw mean percent cover of native and non-native vegetation over time, with ± 1 standard error ribbons around each line and faint transect-level points showing the underlying observations. Cover categories are labeled directly on the figure with `ggrepel` to reduce the processing load of matching a legend. This is distinct from the monitoring report because it uses a continuous line geometry that emphasizes temporal trend, places both cover categories on the same axis for direct visual comparison, and includes a measure of uncertainty that the monitoring report's bar charts do not show.

Figure 3: Dominant non-native species percent cover across survey years

```

# Creating a red/coral color palette for the top non-native species;
# using a single hue family (red to light coral) distinguishes these species
↪ from
# the green native palette while keeping all non-native species visually
↪ grouped
nonnative_palette <- c("#7F1D1D", "#B8322A", "#D85A30", "#E9825A", "#F3B49C")

# Creating Figure 3: top non-native species cover across survey years
fig_3 <- ggplot(data = top_nonnative_by_year,
               aes(x = year,
                   y = mean_pc,
                   fill = species_label)) +
  # Stacked bars show combined non-native cover and each species'
  ↪ contribution
  geom_col(position = "stack") +
  scale_fill_manual(values = nonnative_palette) +
  scale_x_continuous(breaks = seq(2018, 2025, 1)) +
  labs(x = "Survey year",
       y = "Mean percent cover (%)",
       fill = "Non-native species") +
  theme_minimal() +
  theme(legend.position = "bottom",
        legend.text = element_text(face = "italic"),
        legend.title = element_text(face = "bold"))

# displaying figure 3
fig_3

```

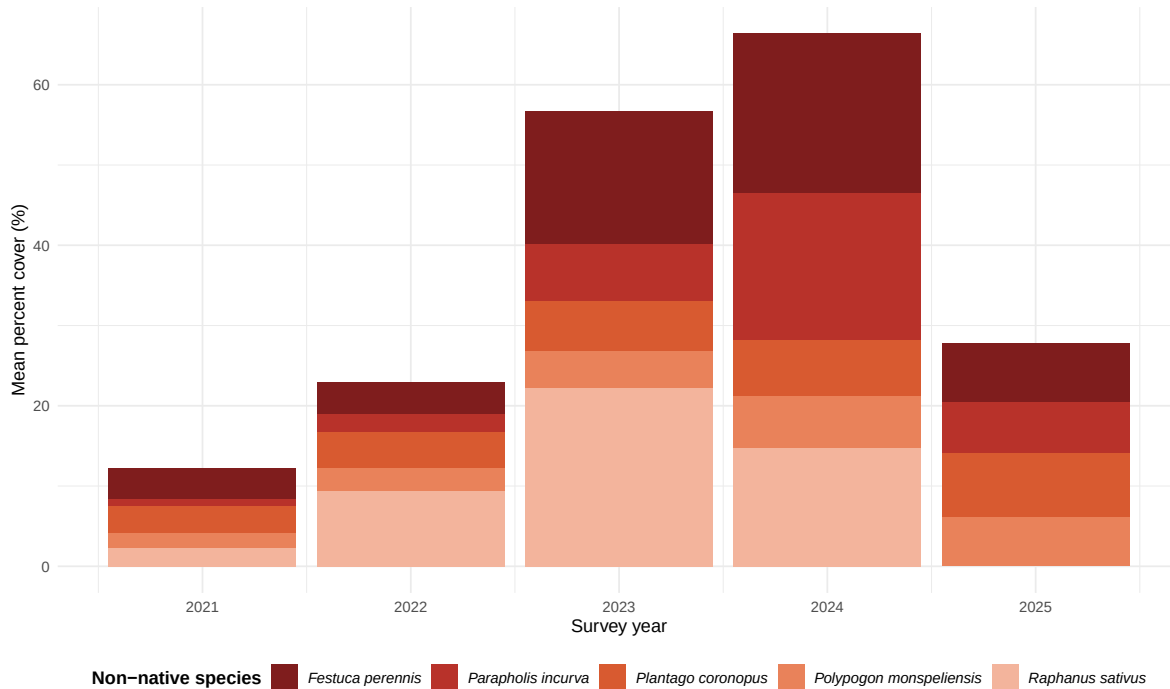


Figure 3: Mean percent cover of the five most common non-native plant species in the filtered salt marsh/brackish marsh dataset across survey years, averaged across transects within each year. Bar height reflects the summed mean cover of these non-native species in each year. Species were selected as the five with the highest mean percent cover across all survey years.

Figure 4: Species rank plot

```
# Creating a transect-year level dataset for underlying raw points in Figure
↪ 4.
# Each point represents one transect-year mean cover value for that species,
# showing the distribution of observations behind the overall mean.
species_rank_raw <- saltmarsh_plants |>
  group_by(psoc, species_label, cover_category, year, transect_name) |>
  summarize(mean_pc_obs = mean(mean_pc, na.rm = TRUE)) |>
  ungroup() |>
  # Keeping only species that appear in the top 20
  filter(psoc %in% species_rank_top20$psoc)

# Setting the x-axis location of the dashed cutoff line from the
# 90th percentile calculated in the dominant-species-cutoff chunk
cutoff_x <- dominance_cutoff
```



```

fig4 <- ggplot(data = species_rank_top20,
               aes(x = mean_pc,
                   y = reorder(species_label, mean_pc),
                   color = cover_category)) +
  # Faint raw points show the spread of transect-year observations for each
  ↪ species,
  # so the reader can assess how variable cover is around the mean
  geom_point(data = species_rank_raw,
             aes(x = mean_pc_obs,
                 y = reorder(species_label, mean_pc_obs)),
             inherit.aes = FALSE,
             alpha = 0.2,
             size = 1.5,
             color = "gray50") +
  # Pointrange shows the overall mean and ± 1 SE for each species
  geom_pointrange(aes(xmin = mean_pc - se_pc,
                     xmax = mean_pc + se_pc)) +
  # Dashed vertical line marks the 90th percentile dominance cutoff;
  # species to the right of this line were selected as dominant for Figure 1
  geom_vline(xintercept = cutoff_x,
             linetype = "dashed",
             color = "gray50") +
  scale_color_manual(values = c("NATIVE COVER" = "#1D9E75",
                                "NON-NATIVE COVER" = "#D85A30")) +
  labs(x = "Mean percent cover (± 1 SE)",
       y = "Species",
       color = "Cover category") +
  theme_minimal() +
  theme(axis.text.y = element_text(face = "italic", size = 9),
        legend.position = "bottom",
        legend.title = element_text(face = "bold"))

# Displaying figure 4
fig4

```

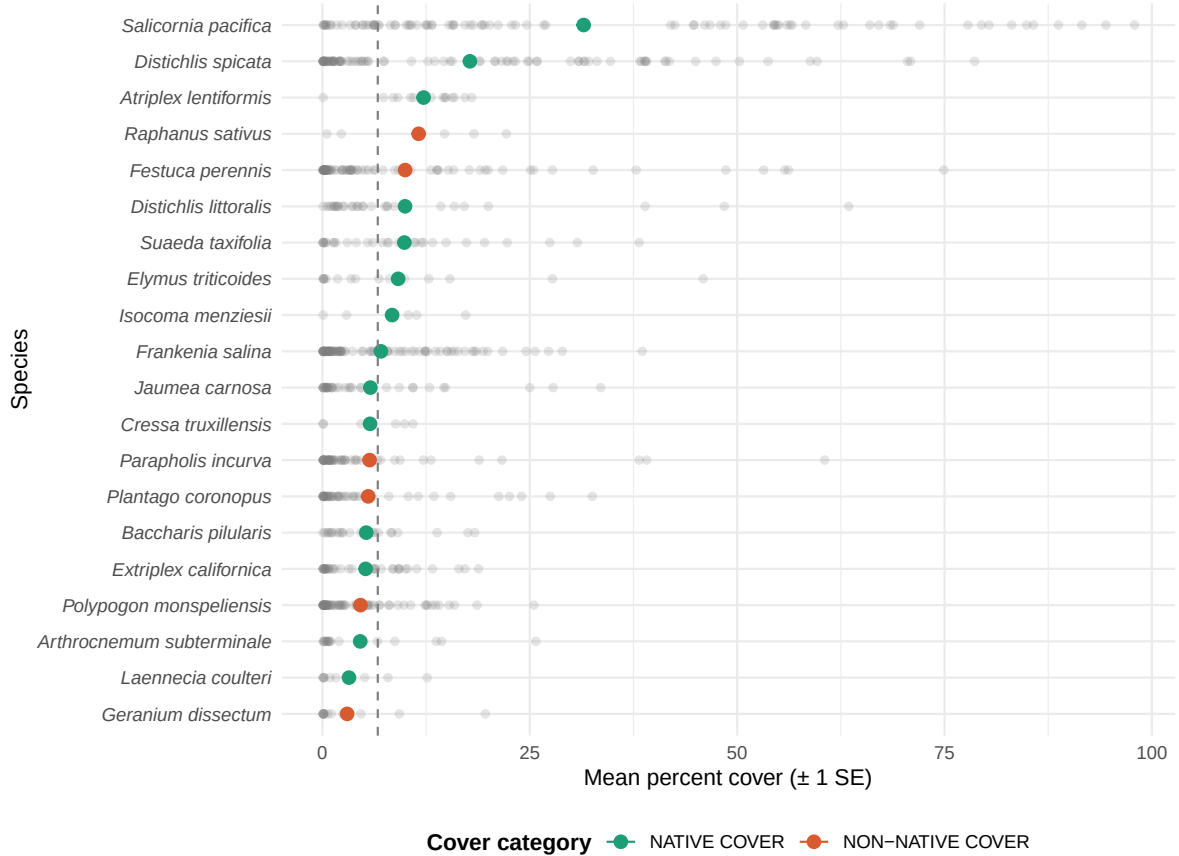


Figure 4: Mean percent cover (± 1 SE) of the top 20 plant species recorded in the filtered salt marsh/brackish marsh dataset, averaged across transects and survey years and ranked from highest to lowest mean cover. Faint points show individual transect-year observations for each species. Points on the pointrange show means; horizontal bars show ± 1 standard error. Green indicates native species; coral indicates non-native species. Dashed line marks the 90th percentile cutoff used to identify dominant species for Figure 1.

The dashed line represents the 90th percentile of mean percent cover across all species in the filtered dataset. Species to the right of this line were classified as dominant because their average cover exceeds that of at least 90% of all recorded plant species. This gives the dominant-species selection a clear mathematical basis rather than relying on a visual cutoff chosen by eye. Faint gray points behind each pointrange show the individual transect-year observations, illustrating how variable cover is for each species across sampling events.

Other exploratory visualizations

Figure 5: Total vegetation cover across survey years

```
# Creating Figure 5: total vegetation cover (native + non-native) across
  ↳ survey years.
# Using a warm amber (#C4820A) rather than the green or coral used for
  ↳ individual
# cover categories, because this figure combines both groups and should not
  ↳ visually
# suggest association with either native or non-native cover alone.

fig5 <- ggplot(data = total_cover,
               aes(x = year,
                   y = mean_total)) +
  # Faint jittered points show the transect-level total cover observations;
  # each point is one transect-year's combined native + non-native cover
  geom_point(data = total_cover_points,
             aes(x = year,
                 y = total_pc),
             inherit.aes = FALSE,
             alpha = 0.35,
             size = 1.8,
             position = position_jitter(width = 0.08, height = 0),
             # Warm amber distinguishes total cover from native (green) and
             ↳ non-native (coral) used in other figures
             color = "#C4820A") +
  # Ribbon shows ± 1 SE around the yearly mean
  geom_ribbon(aes(ymin = mean_total - se_total,
                 ymax = mean_total + se_total),
             fill = "#C4820A",
             alpha = 0.2) +
  # Line connects yearly mean total cover values
  geom_line(color = "#C4820A",
            linewidth = 1) +
  # Larger points mark the yearly means
  geom_point(color = "#C4820A",
            size = 3) +
  labs(x = "Survey year",
       y = "Total mean percent cover (%)") +
  theme_minimal()

# Displaying figure 5
```

fig5

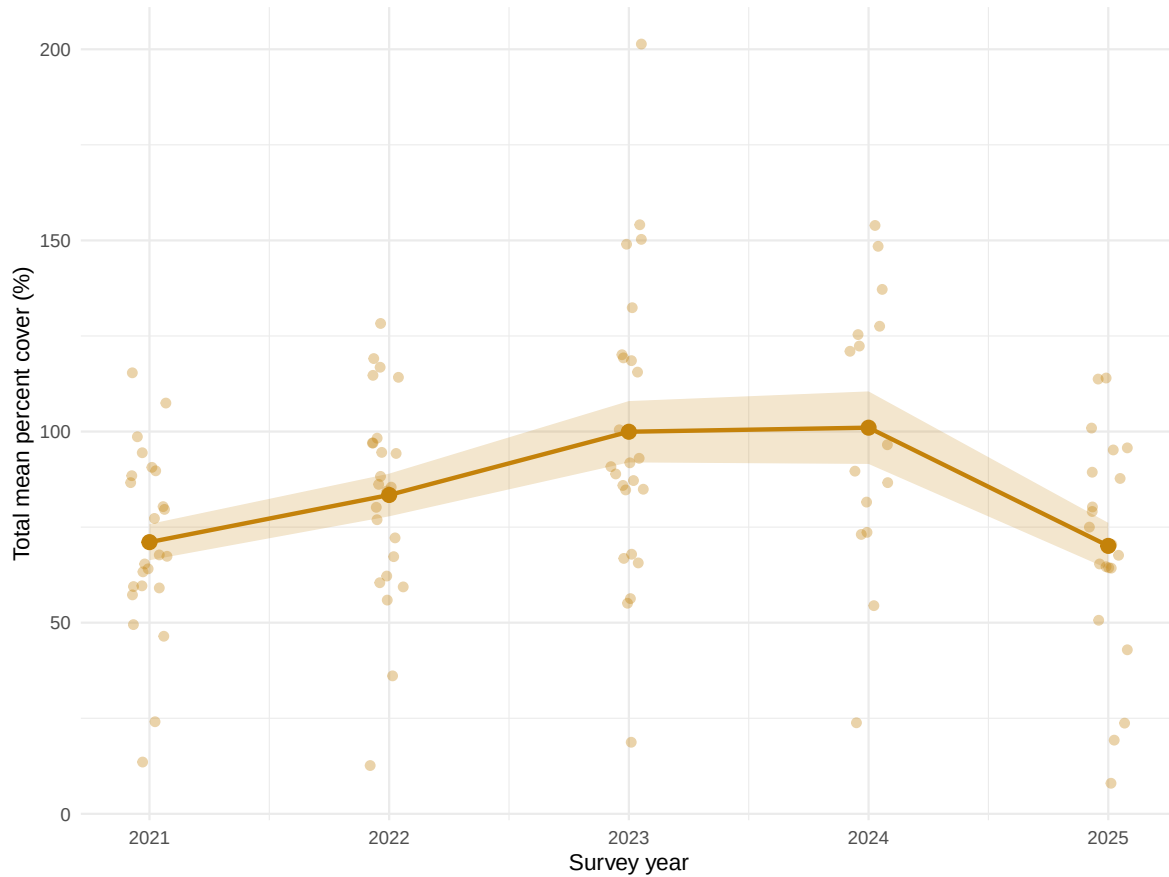


Figure 5: Total mean percent vegetation cover (native + non-native combined) in salt marsh habitat at NCOS across survey years. Faint points show transect-level observations (one point per transect-year); the line connects yearly means; the shaded band represents ± 1 standard error around the mean. A warm amber color is used because this figure combines native and non-native cover rather than representing either group alone.

Plan for elective (if completing)

For our advanced elective, we are creating a National Geographic-inspired educational magazine about the salt marsh plants of the North Campus Open Space (NCOS). The magazine will translate our vegetation analysis into a more accessible and visual format for younger students

and community members, introducing readers to the dominant plant species that characterize salt marsh habitat at NCOS while explaining why salt marshes matter for restoration, wildlife habitat, and coastal resilience. The magazine will be designed in Canva, printed in color at the dorm print facilities, and delivered to Ms. McCarter's fourth grade classroom with one copy per table group (approximately 3–4 copies total).

Elective concept

The magazine is inspired by educational nature publications such as *National Geographic Kids* and is designed for a fourth grade audience. It will include species spotlight pages profiling the dominant salt marsh plants with illustrated profiles, fun facts, and native/non-native status; a “Meet the Marsh” illustrated map of NCOS showing habitat zones; a data feature communicating our project findings in an accessible visual format; a “Native vs. Non-Native vs. Invasive” explanation page; a salt marsh vocabulary word search; a “What Plant Are You?” personality flowchart; and a how-to-get-involved page describing how students can visit and learn more about NCOS.

Draft sections (include some description/image of drafts)

We have finalized the magazine structure, section outline, and visual style. Based on our preliminary data exploration, we have identified the dominant salt marsh species to feature: *Salicornia pacifica*, *Distichlis spicata*, *Atriplex lentiformis*, and *Raphanus sativus*. Each species will receive a spotlight page with a drawing or photograph, its native or non-native status, and a brief description of its role in the marsh community. For images, we will use Creative Commons licensed photographs from iNaturalist, Wikimedia Commons, and the USDA PLANTS Database, supplement with our own photographs taken at NCOS, and create original botanical-style illustrations for species that are difficult to photograph clearly.



Figure 6: Cover page draft



Figure 7: Table of contents draft

ISSUE 01 /

TABLE OF CONTENTS



04.

ABOUT

05.

IMPORTANT
DEFINITIONS

06.

MEET THE MARSH:
A MAP OF NCOS

07

NATIVE VS NON-NATIVE VS
INVASIVE

08.-09.

ECO WORD SEARCH

10.

THE DATA

11.

ENVIRONMENTAL
CROSSWORD

12-13

WHAT PLANT ARE YOU?

14.

SPECIES SPOTLIGHTS

15.

HOW TO VISIT

Figure 8: Table of contents draft 2

ABOUT



Educate, Inspire, Excite

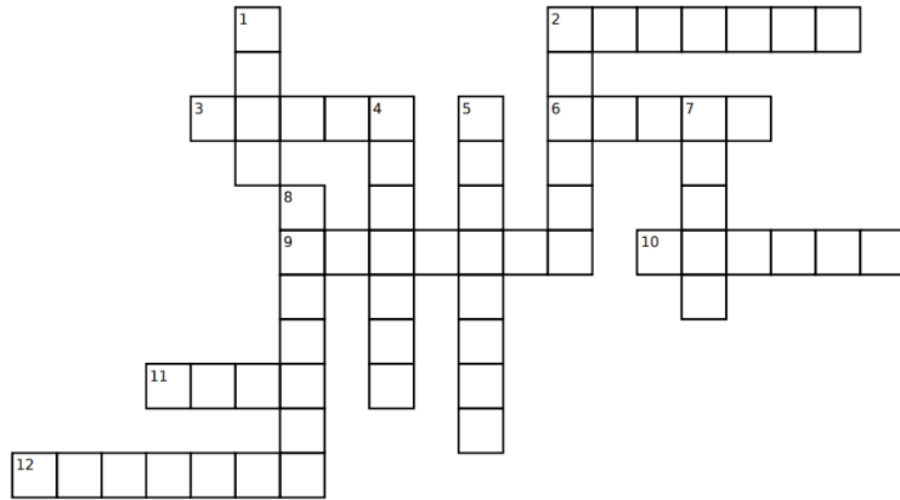
Eco-Explorers Magazine is a children's magazine created for the students of Santa Barbara and Goleta. This edition dives into the salt marsh habitat of the North Campus Open Space (NCOS) — a restored wetland right in our own backyard — and the remarkable plants that call it home. From dominant native species like *Salicornia pacifica* and *Distichlis spicata* to non-native invaders, readers will discover who lives in the marsh, why it matters, and how they can help protect it. Through species spotlights, games, maps, and activities, this magazine aims to educate, inspire, and excite the next generation of environmental stewards.

/ ECO-EXPLORERS

Figure 9: About page draft



Figure 10: Word search draft



CONSERVATION CROSSWORD

Down:

1. Abbreviation for the restored wetland near UCSB (abbr.)
2. Scientists count plants by doing this in the field
4. To bring a habitat back to its natural state
5. A non-native species that spreads and harms the ecosystem
7. Caused by the rise and fall of the ocean
8. A habitat where the ground is soaked with water

Across:

2. A group of plants or animals of the same kind
3. Scientists measure plant ___ to see how much ground is filled
6. The underground parts of a plant that absorb water
9. Where a river meets the ocean, creating a mix of fresh and salt water
10. A plant that naturally belongs in an ecosystem
11. Information scientists collect during a study
12. Someone who protects and cares for the environment

/ECO-EXPLORERS

Figure 11: Crossword draft

Plan for the next two weeks

Over the next two weeks, we will use our preliminary analysis to select the species featured in the magazine. We will also continue revising our figures so the main results are clear enough to translate into a more visual and accessible format for younger students and community readers.

By Week 9, we will have completed all remaining sections including the data feature, word search, flowchart, and how-to-get-involved page, and will have begun design and layout in Canva. By Week 10 we will finalize and polish the full magazine for presentation, revising the writing for clarity and ensuring the layout is cohesive and visually engaging. The completed magazine will be delivered to Ms. McCarter’s classroom during finals week.

Next steps

Our next steps are to refine the statistical analyses and continue revising the figures so our interpretations are clear and well-supported. For the final project, we will add more complete citations, connect our results more directly to NCOS restoration goals, and continue developing the advanced elective magazine alongside the data analysis.

Before submission, we will make sure the GitHub repository includes the rendered PDF, the QMD, the data files, the bibliography file, and a README that clearly links to the rendered document.

References

- Beheshti, Kathryn M., Stephen C. Schroeter, Andres A. Deza, Daniel C. Reed, Rachel S. Smith, and Henry M. Page. 2023. “Large-Scale Field Studies Inform Adaptive Management of California Wetland Restoration.” *Restoration Ecology* 31 (5): e13936. <https://doi.org/10.1111/rec.13936>.
- Cheadle Center for Biodiversity and Ecological Restoration, and Environmental Science Associates. 2016. “North Campus Open Space Restoration Project Restoration Plan.” University of California, Santa Barbara. <https://escholarship.org/uc/item/9505h60h>.
- Ferren, Wayne R. Jr. 1985. *Carpinteria Salt Marsh: Environment, History, and Botanical Resources of a Southern California Estuary*. Publication Number 4. Santa Barbara, CA: The Herbarium, Department of Biological Sciences, University of California, Santa Barbara.
- North Campus Open Space. 2025. “Coastal Salt Marsh.” <https://www.ncos.ccber.ucsb.edu/native-plant-habitats/coastal-salt-marsh>.
- Rickard, Alison. 2024. “North Campus Open Space Restoration Project Monitoring Report: Year 7, December 2024.” University of California, Santa Barbara Cheadle Center for Biodiversity; Ecological Restoration. <https://escholarship.org/uc/item/7bq618m8>.